Weil Classes on Abelian Powers: Finite Supply Lattices and Exact Weil-Plane Certificates

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Abstract

We examine the tensor and cycle-supply interfaces in *Weil classes and Hodge classes on abelian powers*, without claiming an independent proof of its geometric or companion CM inputs. For fixed admissible blocks at the actual period, we replace the determinant criterion's unbounded repetition condition by finitely many supply obligations. A Hilbert-basis formulation keeps all positive lists; a stronger signed-lattice formulation uses opposite determinants and nonzero polarization contractions, reducing the linear-block obligations to at most their number. Integral witnesses, not just rational nullspaces, are retained. We also construct a rational algebraic projector onto the complete imaginary-quadratic Weil plane by a polynomial of minimal degree in the square of the infinitesimal field action. On a sixfold this polynomial is explicit and yields an automatically independent, optimally balanced two-column certificate from one nonzero projected cycle. We prove the reductions, give exact examples and failure tests, and state finite-moment error budgets. No numerical rank establishes the seed, the full obstruction calculation, period descent, or the companion CM Hodge theorem. Contribution assessment: C3/5, a notable local reduction of source-specific supply obligations and a constructive whole-Weil-plane certificate, conditional on genuine geometric inputs.

Contents

1 Scope, Source, and Literature **2**

2 Exact Spaces and Geometric Gates **3**

3 A Finite Positive-List Criterion **4**

4 Signed Supply Lattices **4**
4.1 An exact finite implementation contract 6
4.2 The source's special-period cases 6

5 Localizing the CM Hodge Input **6**

6 An Explicit Projector onto the Whole Weil Plane **7**

7 One Projected Cycle, Two Certified Directions **8**
7.1 A complete period error budget 9
7.2 Full Hodge space versus the Weil plane 10

8 Checks, Retained Details, and Limitations **10**

9 Contribution Assessment **11**

1 Scope, Source, and Literature

The applicability rating remains L3. The independent contribution score concerns the results proved here, not the truth of the source's main claims and not the number of exact tests. The source [1] asserts the rational Hodge conjecture in every codimension of every self-power in two settings: a sixfold with an imaginary-quadratic action and an actual compatible hyperbolic Hermitian polarization of signature $(3, 3)$; and a variety of dimension at most five with an imaginary-quadratic action. Special periods and nonsimple varieties are included in those assertions. They are not independently certified here.

We literally read the entire introductory, tensor, and actual-period reduction, source lines 1–1884, together with the exact later intervals recorded in the owned read ledger. The source is one 8285-line file. The remaining geometric proof is mapped by section and theorem locators, not represented as fully read. The principal additional inspected passages are the full rank-comparison argument, the local extraction, the two-generator lemma, and the final isogeny argument; selected seed, spin, obstruction, deformation, and spread statements identify their precise roles. Literal reading is itself weaker than an independent proof audit. Our new deductions do not use unread geometric arguments as if they had been checked.

Source architecture	Role and review boundary
Lines 240–804	CM supply, admissible operations, all-degree tensor criterion, and transfer. Read throughout.
805–1835	Actual Albert columns, independent derived projections, and simultaneous balances. Read throughout; classification and geometric supply remain imported when drawing global conclusions.
1836–3026	Abel-orbit seed, actual coherent transform, finite action. Mapped; selected seed statement only.
3027–4831	HKR/Atiyah identities and full invariant derived obstruction. Selected statements and full final rank comparison, not the entire proof.
4832–5705	Fixed spin identity, actual polarization, exceptional character. Selected statements; no independent spin calculation.
5706–7159	Twisted descent, ordinary conversion, semiregularity. Selected deformation theorem; geometric hypotheses imported.
7160–8024	Marked domain, local extraction, full-plane generation, Hilbert spread, lattice isogeny. Selected literal intervals only.

The October 8 primary-literature refresh distinguishes several statements. Markman's current survey, Theorem 1.2 and Corollary 1.3, states algebraicity of the Weil classes on imaginary-quadratic Weil fourfolds and split Weil sixfolds, and the Hodge conjecture for individual abelian varieties of dimension at most five [2]. These statements do not alone identify all tensors on all their powers at special periods. Floccari–Fu, Theorem 1.2, explicitly states the all-powers result for Weil fourfolds of

discriminant one [3]. We inspected these precise statements, not their complete proofs. Schrijver's mixed-tensor first fundamental theorem supplies pair and determinant generators for SL invariants [4, Section 5]; this is classical input, not a new Ram result. Gordan finite generation is likewise classical [5, Theorem 2]. No general algorithm in those references produces the missing algebraic cycles.

2 Exact Spaces and Geometric Gates

Fix a complex abelian variety A, $H = H^1(A, \mathbb{Q})$, and $k = \overline{\mathbb{Q}} \subset \mathbb{C}$. Cohomology is singular rational cohomology; $H(r)^{a,b} = H^{a+r,b+r}$, so a twist by r lowers weight by $2r$. Scalar Galois conjugation changes coefficients on this fixed variety, not the variety. All tensor slots are distinct copies of A until an explicit algebraic diagonal pullback is made. Exterior and ordinary tensor permutations differ by the usual Koszul signs; no unrecorded division by a zero contraction is permitted.

The source calls a subspace CM-supplied if it is contained in

$$\mathcal{C}^d(X)_k = \left(\sum_{B,j,\Gamma} \operatorname{im}\{\Gamma_* : H^{d-2j}(B, \mathbb{Q})(-j) \longrightarrow H^d(X, \mathbb{Q})\} \right)_k, \tag{1}$$

where B is an actual CM abelian variety and $\Gamma \in \mathrm{CH}^{\dim B+j}(B \times X)_{\mathbb{Q}}$ an actual correspondence. The displayed twist states the Hodge-map convention; the underlying cohomological map has degree shift $2j$. Supply uses full CM cohomology, not just its Hodge classes. Products, scalar conjugation, and rational algebraic correspondences preserve this rational subspace. A rational Hodge class in it lifts through a finite sum of these maps by splitting polarizable Hodge structures. Algebraicity of that lift still requires the companion CM Hodge theorem. M061 is a local comparative report and is not independent certification of that theorem.

For an admissible degree-one summand V of dimension r, let $D(V) = \Lambda^r V$ on separate slots, embedded by unnormalized alternation. A pair tensor gives a perfect inverse pairing $b : V \otimes \bar{V} \to k$ of bidegree $(-1, -1)$. If e_i, f_j are dual for b, then

$$b^{\otimes r}\left(\mathrm{Alt}(e_1 \otimes \cdots \otimes e_r) \otimes \mathrm{Alt}(f_1 \otimes \cdots \otimes f_r)\right) = r!, \tag{2}$$

after pairing corresponding slots. For a general pairing matrix the answer is $r! \det(b(e_i, f_j))$. Expanding the two permutation sums proves the formula: for each first permutation the second signed sum is its sign times the determinant. The cohomological ordering may introduce a fixed sign, never zero. Normalized alternation changes the factor to $\det(b)/r!$. We use the unnormalized convention throughout. The pairing and slot maps must be admissible algebraic operations, as in the source, not arbitrary linear identifications.

Assume the source criterion's actual-period block data: a toric summand H_0 with fixed CM embedding lines; symplectic blocks; orthogonal blocks with all-conjugate symmetric pair tensors; and linear blocks $V_i, \bar{V}_i$, $1 \le i \le s$, with their admissible identifications. The actual derived image must contain the independent product of the corresponding classical groups. Projection onto each group separately does not imply this product condition. Pairings already supply the symplectic generators. Orthogonal volumes must be supplied separately; their imbalances vanish and their ranks are even.

For any selected summand define

$$\epsilon_\gamma(V) = \dim(\gamma V)^{1,0} - \dim(\gamma V)^{0,1}.$$

The source proves $\epsilon_\gamma(\bar{V}) = -\epsilon_\gamma(V)$ for every γ, without assuming it commutes with conjugation. Fix a finite Galois coefficient field containing every projector, identification, and embedding line. Restrictions of γ to this field give finitely many balance rows. Keeping all of them, or an exactly proved equivalent row system, is essential. The choice of CM lines is fixed before any balance test.

3 A Finite Positive-List Criterion

List all oriented determinant types and all actual CM line types. Let $\mathsf{A} \in \mathbb{Z}^{h\times q}$ have their complete imbalance vectors as columns and put

$$S_{\mathsf{A}} = \{a \in \mathbb{N}^q : \mathsf{A}a = 0\}, \qquad \mathbb{N} = \{0, 1, \ldots\}.$$

Multiplicity variables count factors, not coefficients of their tensor vectors. Each column also has a positive tensor degree: a determinant's rank or one for a CM line. In each row its imbalance is congruent to that degree modulo two. Thus every balanced list has even total degree. The balance equation contains no missing parity restriction.

Proposition 3.1 (Finite positive obligations)**.** *Let $\mathcal{H}$ be a complete Hilbert basis of S_{A}. Under the block hypotheses above, the source's condition (d) is equivalent to CM supply of the determinant part of each $a \in \mathcal{H}$.*

Proof. Here is a direct finiteness argument. Every infinite sequence in $\mathbb{N}^q$ contains a comparable pair in coordinatewise order: induct on q, passing in the first coordinate either to an infinite constant subsequence or to a strictly increasing subsequence, and apply induction to the remaining coordinates. Hence a subset of $\mathbb{N}^q$ has finitely many minimal elements. Apply this to $S_{\mathsf{A}} \setminus \{0\}$. If $0 < b \le a$ are balanced, then $a - b$ is balanced. Its coordinate sum is smaller unless $a = b$. Induction decomposes every balanced vector into minimal nonzero balanced vectors. These are precisely the Hilbert atoms. This also proves their uniqueness as a minimal generating set.

Necessity is immediate from (d). For sufficiency decompose the list vector into atoms, take external products of the supplied determinant parts, and permute their slots to match the original list. All CM entries were used in the same simultaneous equation; none was chosen afresh for a different conjugate. Slot permutations and admissible copy identifications preserve supply. This proves (d) in every degree. □

The positive method is useful if one wants a certificate using only products, without cancellation. It requires a *complete* integer Hilbert basis, not a finite enumeration guessed to have stabilized. For $\mathsf{A} = (1, 1, -2)$ the atoms are $(2, 0, 1), (1, 1, 1), (0, 2, 1)$. To prove completeness, if both first coordinates are positive subtract $(1, 1, 1)$; otherwise the nonzero coordinate is even and a multiple of the corresponding outer atom. The two primitive extreme rays miss the middle atom. Degrees $(3, 3, 2)$ give compatible parity, and the middle atom has even degree eight. This is formal balance data, not a constructed abelian variety.

4 Signed Supply Lattices

The source has additional structure beyond positive monoid closure. Choose one line from each opposite CM embedding pair, denoted $\ell_1, \ldots, \ell_t$. This is possible using rational rank-one CM module decompositions; both orientations remain available on arbitrary repeated slots. Write

$$A_0 = (\epsilon_\gamma(V_i))_{\gamma,i}, \quad B_0 = (\epsilon_\gamma(\ell_j))_{\gamma,j}, \quad L = \{n \in \mathbb{Z}^s : \exists m \in \mathbb{Z}^t, A_0 n + B_0 m = 0\}. \tag{3}$$

When no CM lines are present $L = \ker_{\mathbb{Z}} A_0$. Define

$$D(n) = \bigotimes_i D(V_i)^{\otimes n_i^+} \otimes D(\bar{V}_i)^{\otimes n_i^-}, \qquad n_i^+ = \max(n_i, 0), \quad n_i^- = \max(-n_i, 0).$$

Every negative witness entry is realized by the corresponding opposite CM line. Thus an integer witness is a genuine positive list after orienting factors; it is not a formal negative tensor degree.

Lemma 4.1 (Supply is a subgroup)**.** *The set $\{n \in \mathbb{Z}^s : D(n)$ is CM-supplied$\}$ is a subgroup. Moreover an arbitrary positive determinant list of net exponent n is supplied if and only if $D(n)$ is supplied.*

Proof. The unit line is supplied by a point. Scalar complex conjugation sends $D(n)$ to $D(-n)$, up to the specified admissible copies; the supply space is rational and hence stable under this operation. Tensor supplied $D(n)$ and $D(n')$. Every opposing pair can be contracted by (2). Each contraction is nonzero, and the resulting one-dimensional image is exactly $D(n + n')$. The Tate degree shifts are those of the polarization contraction. This proves closure under addition and inverses.

For the final assertion, contract all opposite pairs in the original list to obtain $D(n)$. Conversely each deleted pair is supplied by alternating tensor powers of the algebraic coevaluation. Tensor these back with $D(n)$ and reorder slots. In neither direction do we take a root of a tensor or infer a factor's supply from its power. □

Theorem 4.2 (Finite lattice-basis criterion)**.** *Retain the source's block hypotheses (a)–(c), the complete balance rows, and the fixed opposite-paired CM list. Let $b_1, \ldots, b_r$ be a $\mathbb{Z}$-basis of the projected lattice L in* (3)*. Then condition (d) is equivalent to the following finite obligations: every orthogonal singleton volume is CM-supplied, and each $D(b_j)$ is CM-supplied. In particular there are at most s linear-block obligations. With the source's companion CM theorem, these obligations imply its all-powers conclusion; without that theorem they only give the stated CM-supply conclusion for the balanced tensor generators.*

Proof. An orthogonal singleton is balanced in every conjugate and is among the lists in (d). Each b_j has an integral witness m_j. Realize its positive and negative entries using the fixed orientations of determinants and CM lines. The complete balance equations hold, so (d) supplies $D(b_j)$. This proves necessity.

Conversely take an arbitrary list in (d). Remove its orthogonal entries, which are supplied and have zero imbalance. Cancel its opposite linear determinant pairs and opposite CM pairs. The resulting net determinant vector n and CM vector m obey (3), so $n = \sum c_j b_j$ with $c_j \in \mathbb{Z}$. The subgroup lemma supplies $D(n)$ using $|c_j|$ copies of the appropriate orientation of $D(b_j)$. Restore the deleted supplied pairs and orthogonal factors. This gives precisely the original determinant list. The source's classical invariant spanning, simultaneous-balance projection, rational Hodge lift, and companion CM input give its subsequent conclusion under their hypotheses. No assertion that these hypotheses hold for a new variety is made here. □

This is a source-specific strengthening of Proposition 3.1: the relevant supply closure is a group because actual pairings allow cancellation. Finiteness of subgroups of $\mathbb{Z}^s$ is classical. The gain is in the *geometric obligations being tested*, not a new integer normal-form theorem. It differs from M061's batching of fixed CM-type switch paths; no such switching result is used here.

4.1 An exact finite implementation contract

Put $C = [A_0\ B_0]$. A certified Smith decomposition $UCV = S$ has integer unimodular U, V, diagonal nonzero entries followed by zeros, and exact equality. The columns of V in the zero-diagonal positions form an integer basis of $\ker_{\mathbb{Z}} C$: writing $x = Vy$, the equation $Sy = 0$ forces exactly its nonzero-diagonal coordinates to vanish. Project these columns to the first s coordinates. Their integer column span is L. A column-Hermite reduction, with its exact integer change-of-generator records, extracts a basis. Preserve the last t coordinates through those operations to retain witnesses for each b_j. Verify unimodularity and both equalities exactly. Rank alone is insufficient.

Neither a rational kernel basis with individually cleared denominators nor saturation of the projected lattice is a substitute. For the formal system

$$B_0 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \qquad A_0 = \begin{pmatrix} 2 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \tag{4}$$

$B_0^T B_0 = 4I$, so its unique rational witness is $m = -n(1,1,1,1)^T/2$. Hence $L = 2\mathbb{Z}$, although the projected rational solution space is all of $\mathbb{Q}$. Append negative rows and opposite columns if desired; the condition is unchanged. The determinant column has degree two and each CM column degree one, so all row parity conditions hold. This is a manufactured integer-balance example, not a claim of geometric realization of its complete Galois data. Supply of $D(2)$ does not give supply of $D(1)$ by tensor-root extraction.

4.2 The source's special-period cases

For two higher-center columns and a CM contribution fixed over E, the two equations are $d_1 n_1 + d_2 n_2 + c = 0$ and $d_2 n_1 + d_1 n_2 + c = 0$. If $d_1 \neq d_2$, subtraction gives $n_1 = n_2$. For $(d_1, d_2) = (0, 2)$ with an available E-CM line, the projected lattice is $\mathbb{Z}(1,1)$; a witness has net CM exponent -2 per copy. Without that CM line both equations force $n_1 = n_2 = 0$. Thus removing CM lines can change the obligations. For $(-1, 1)$ with no CM remainder, $L = \mathbb{Z}(1,1)$. For the three cyclic rows of $(-2, 0, 2)$, the equations force $L = \mathbb{Z}(1,1,1)$, with no hidden index. These are exactly the source's higher-center reductions. One full E-determinant supply obligation suffices in each nontrivial case, not one per column.

In the formal example $(1, 1, -2)$, a signed lattice basis is $(1, -1, 0), (1, 1, 1)$: for $n_1 + n_2 = 2n_3$, the coefficients are $(n_1 - n_2)/2$ and n_3, which are integers. Two signed obligations replace three Hilbert atoms, if all opposite lines and contractions are available. Conversely checking only the two extreme rays leaves a subgroup of index two and still misses $(1, 1, 1)$.

The number of obligations is not a bound on cycle degree, cover degree, height, or time to find a cycle. A lattice basis may have large entries, and negative combinations may consume many supplied factors. For $n = \sum c_j b_j$, a transparent tensor-slot budget before cancellation is $\sum_j |c_j| \sum_i (\dim V_i)|(b_j)_i|$; the final determinant degree is $\sum_i (\dim V_i)|n_i|$. The number of paired slots contracted is half their difference. These are algebraic-operation counts, not geometric complexity estimates.

5 Localizing the CM Hodge Input

Theorem 5.1 (One finite CM library)**.** *Fix A and the data of Theorem 4.2. Suppose finite correspondence witnesses for its finitely many supply obligations have actually been specified. There*

is one CM abelian variety B_*, a product of those witnesses' CM sources and a CM isogeny factor supplying H_0, such that every rational balanced tensor is supplied from finite products of members of this library. It is sufficient to assume the rational Hodge conjecture on every power of this one B_* to deduce the all-powers conclusion for A. The same assertion holds with a fixed additional CM companion after adjoining that companion to B_*.

Proof. Finite dimensionality and the definition (1) select finitely many correspondence maps for each of the finitely many lines. Expanding their scalar coefficients leaves finitely many rational correspondence cycles. List their CM sources $B_1, \dots, B_v$ and the CM factor for H_0. Form their product B_*. Pair tensors use a point as source. Conjugating a supplied line does not require conjugating its underlying variety: the sum of full images of its rational correspondence maps is rational, so the conjugate line lies in the same sum.

The constructive proof of Theorem 4.2 now uses only external products of these sources followed by algebraic slot maps and contractions. In any fixed tensor degree, only finitely many such maps are needed. The invariant spanning theorem and complete balance test put the target rational tensor in their rational image sum. The polarized splitting gives a rational Hodge lift on a finite direct sum of cohomologies of library products, with the recorded Tate twists. Every such product is an abelian factor of a sufficiently high power of B_*. Its Hodge classes pull back to that power and return by an algebraic factor inclusion at the origin. Thus the assumed Hodge conjecture on all powers of B_* algebraizes every component of the lift. Applying the original correspondences gives the target cycle. Repeated A slots only reuse the same library. Adjoining a fixed CM companion adds its degree-one source to the product and changes no step. □

The library is conditional on actual supply witnesses. No finite list of numerical periods constructs them. The theorem does not replace Hodge on all powers by Hodge on each B_j individually: mixed Hodge classes on products are part of the hypothesis. Nor is a finite bound on the power proved for the simultaneous conclusion in every degree. This isolates precisely how much of the companion theorem a fixed application needs, without claiming a new CM case.

6 An Explicit Projector onto the Whole Weil Plane

Let $E = \mathbb{Q}(u)$, $u^2 = -d$, $d > 0$ rational. Let A have dimension $N = 2n$ and a unital E-action. Put $H = H^1(A, \mathbb{Q})$; its E-rank is N. The compatible balanced signature (n, n) makes

$$W_E(A)_k = \Lambda^N H_\sigma \oplus \Lambda^N H_{\bar\sigma} \subset \Lambda^N H_k = H^N(A, k)$$

a rational two-dimensional Hodge subspace of type (n, n). Its dimension remains two at special periods, even when the full Hodge space becomes larger. The projector below exists also without balance, but its image is then not claimed to be a Hodge-class space.

Write $U = u^*|_H$, so $U^2 = -dI$. Define on $\Lambda^N H$

$$T(v_1 \wedge \cdots \wedge v_N) = \sum_{i=1}^{N} v_1 \wedge \cdots \wedge Uv_i \wedge \cdots \wedge v_N. \tag{5}$$

This is the infinitesimal action, not u^* on degree N: the latter applies U to every slot at once. In particular for even N its two endpoint eigenvalues agree and it cannot by itself separate the plane.

Lemma 6.1 (Algebraic realization)**.** *The rational operator T on $H^N(A, \mathbb{Q})$ is induced by a rational algebraic correspondence. It can be implemented using $N + 1$ rational endomorphism pullbacks and exact coefficient interpolation.*

Proof. On degree N, the pullback by the rational endomorphism $z+u$ is

$$(z+u)^*|_{H^N} = \Lambda^N(zI+U).$$

It is a polynomial of degree N in z whose coefficient of z^{N-1} is exactly (5). Evaluate at $z = 0, 1, \ldots, N$. The rational Vandermonde inverse extracts that coefficient. If a_z clears the denominator of $z+u \in \mathrm{End}^0(A)$, the corresponding rational pullback on H^N is a_z^{-N} times the pullback by the genuine algebraic endomorphism $a_z(z+u)$. Thus each interpolated operator is algebraic. If an operator on full cohomology is wanted, compose with

$$\pi_N = \prod_{\substack{0 \le r \le 2N \\ r \ne N}} \frac{[2]^* - 2^r I}{2^N - 2^r},$$

the rational degree-N correspondence projector. Composition and rational linear combination preserve algebraicity. □

Theorem 6.2 (Folded spectral projector)**.** *On $H^N(A, \mathbb{Q})$ put*

$$\Pi_E = \prod_{j=1}^{n} \frac{T^2 + d(N-2j)^2 I}{d((N-2j)^2 - N^2)}. \tag{6}$$

It is a rational algebraic correspondence projector whose image is exactly $W_E(A)$. Among polynomials in T^2 with this action its degree n is minimal. For a sixfold,

$$\Pi_E = -\frac{T^2(T^2+4dI)(T^2+16dI)}{23040d^3}, \qquad T^2|_{W_E(A)} = -36dI. \tag{7}$$

Proof. Over E, H is the sum of two N-dimensional eigenspaces with eigenvalues $u, -u$. On the summand $\Lambda^j H_\sigma \otimes \Lambda^{N-j} H_{\bar\sigma}$, T acts by $(2j-N)u$, and T^2 by $-d(N-2j)^2$. Each summand is nonzero, of dimension $\binom{N}{j}^2$. The distinct squared eigenvalues are indexed by $0 \le j \le n$. Formula (6) is one on the endpoint value $-dN^2$ and zero on the other n values. Its image is exactly the two endpoint lines; Galois descent makes it their rational plane. Semisimplicity gives idempotence. All coefficients and all composing correspondences are rational. Any polynomial with the required action has the other n distinct eigenvalues as roots and does not vanish at the endpoint, so has degree at least n. For $N = 6$ the denominator is $d^3(-20)(-32)(-36) = -23040d^3$, proving the formula. □

This is a minimal-degree statement only within polynomials in T^2, not a claim of minimal geometric cost among all correspondences. Squaring identifies conjugate eigenvalues, which is exactly appropriate for the rational plane and would be inappropriate for a single complex embedding line. No generic-Hodge-group assumption is used.

7 One Projected Cycle, Two Certified Directions

Theorem 7.1 (Structured full-plane certificate)**.** *Let $Z \in \mathrm{CH}^n(A)_{\mathbb{Q}}$ be a genuine rational algebraic cycle and $w = \Pi_E \,\mathrm{cl}(Z)$. If $w \ne 0$, then w, Tw are rational algebraic classes forming a basis of the whole $W_E(A)$. More generally the same conclusion holds for any nonzero rational algebraic w known exactly to lie in this plane.*

Proof. Theorem 6.2 enforces exact membership by an algebraic correspondence; it does not just discard a small numerical complement. The operator T is algebraic. A rational linear dependence between w and Tw would give $Tw = aw$ for $a \in \mathbb{Q}$, since $w \neq 0$. But $a^2 w = T^2 w = -dN^2 w$, impossible for $d > 0$. The two independent classes span the rational plane. □

The source already proves that one nonzero rational algebraic Weil class generates the plane under a suitable k^*, $k \in E^\times$ (lines 7598–7633). We do not claim that qualitative fact as new. The constructive increment is an exact projector on the full ambient degree, the fixed infinitesimal generator with no exceptional choice of k, and the following quantitative two-direction normalization.

Choose any nonzero rational reference $e \in W_E(A)$, not assumed algebraic, and take the rational basis $e, (T/N)e$. In coordinates,

$$T = N \begin{pmatrix} 0 & -d \\ 1 & 0 \end{pmatrix}, \quad w = \begin{pmatrix} x \\ y \end{pmatrix}, \quad \det[\, w \;\; Tw \,] = N(x^2 + dy^2). \tag{8}$$

Thus the full-rank test is the nonvanishing of one positive quadratic form, not two independently uncertain period columns. In the positive metric $G = \mathrm{diag}(1, d)$, $J = T/(N\sqrt{d})$ is orthogonal and skew-adjoint. Therefore

$$[\, w \;\; Jw \,]^T G [\, w \;\; Jw \,] = (x^2 + dy^2) I_2. \tag{9}$$

The two normalized columns have condition number one. The real scaling $1/(N\sqrt{d})$ is for conditioning only; rational algebraic cycle combinations use w, Tw. A nonzero rational vector always has both complex embedding coordinates nonzero; one complex line alone is not the rational target.

7.1 A complete period error budget

Let B be an injective two-test pairing matrix for the stated rational basis of the plane, and let p be the tests of the actual projected cycle. With outward-rounded bounds $\left\| B - \widehat{B} \right\|_2 \leq e_B$, $\|p - \widehat{p}\|_2 \leq e_p$, $\left\| \widehat{p} - \widehat{B}\widehat{w} \right\|_2 \leq e_s$, and $s_B \leq \sigma_{\min}(\widehat{B})$, $s_B > e_B$, one has

$$\|w - \widehat{w}\|_G \leq \sqrt{\max(1, d)}\, \frac{e_p + e_B \|\widehat{w}\|_2 + e_s}{s_B - e_B} =: \varepsilon_G. \tag{10}$$

Subtract $B\widehat{w}$ from p, insert the approximations, and use $\left\| B^{-1} \right\|_2 \leq (s_B - e_B)^{-1}$ to prove this bound. If a certified lower bound for $\|\widehat{w}\|_G$ exceeds ε_G, then $w \neq 0$ and Theorem 7.1 applies. For the balanced matrix in (9), the error from replacing w by $\widehat{w}$ is exactly $\|w - \widehat{w}\|_G$ in the mass-scaled spectral norm: apply the same Gram identity to their difference. There is no unnecessary $\sqrt{2}$ penalty for two perfectly correlated columns.

If T is implemented with an additional operator error, it must be included; the exact correlation argument then no longer absorbs it. For $\widehat{C} = [\widehat{w}, \widehat{J}\widehat{w}]$, the second-column error is bounded by $\varepsilon_G + \left\| \widehat{J} - J \right\|_G \|\widehat{w}\|_G$ when the first term uses the exact J. This is a deterministic error budget, not permission to substitute floating-point membership for an algebraic projector.

The original private TR-item-011, lines 241–288 and 513–656, already requires actual cycles, exact target membership, rationality, and a rigorous full-span margin. Its original perturbation and lattice-gap proofs were read, not inferred from a catalogue summary. The original TR248 finite-frame proofs distinguish an exact finite model from unmeasured components and require atom/quadrature errors to enter the norm budget. Here those safeguards are specialized to the complete Weil plane and its correlated two-column structure. The source's geometric seed is still the place where a cycle with $\Pi_E \,\mathrm{cl}(Z) \neq 0$ must be produced.

7.2 Full Hodge space versus the Weil plane

A plane certificate is not a full Hodge-space certificate. At a special period, fix the actual rational target V and a certified rational matrix of generator columns K supplied by the actual invariant and balance analysis. For an actual rational cycle matrix C, the exact full-span condition is

$$\operatorname{im} C = V = \operatorname{im} K,$$

with exact membership of both matrices in V. For example a rational left inverse on a basis of $\operatorname{im} K$ and a rational solution of $CR = K$ certify spanning; a rational functional annihilating C but not K certifies a missing direction. Relations among invariant generators must be retained, so counting diagrams is not counting dimensions. The source's complete all-conjugate balance projection remains necessary; the condition at one embedding alone can accept nonrational Hodge directions.

8 Checks, Retained Details, and Limitations

The owned executable suite passed 74,372 exact assertions in 26,585 manufactured cases. It uses integers and `fractions.Fraction`, without floating-point rank decisions. It checks bounded kernel decompositions, signed-basis reconstruction, the nonsaturated witness example, the source's two- and three-column balances, determinant contraction normalization, rational projector interpolation, and the structured two-column identity. Exact polynomial divisibility certifies projector identities against the declared spectral polynomial; small exterior models independently test the infinitesimal action. These are manufactured finite instances, not computations of an actual sixfold's periods, Hilbert schemes, or algebraic cycles. Exact counts and the parameter ranges are reported in the accompanying JSON. Parent tests and reviews are not included in the author-run counts.

No substantive error is asserted in the inspected source reductions. We retain its valid distinctions: the full derived invariant obstruction space rather than only a Hochschild image; exact rank comparison after an exact triangle identity; full CM cohomology versus Hodge classes; noncentral quadratic actions and actual endomorphism columns; no assumption that a summand of a hyperbolic form is hyperbolic; the nonzero determinant contractions; and the whole rational Weil plane rather than one complex eigendirection. The new tests do not certify the source's prior geometric dimension or transport inputs.

Advantages. The signed criterion changes an infinite list of geometric obligations to an exact finite library with witnesses. The positive Hilbert method remains an alternative when cancellation is unavailable. The folded projector enforces membership on the entire ambient cohomology degree, and the structured frame reduces a full-plane rank certificate to one nonzero projected cycle with an explicit error margin. All constructions are invariant under extra endomorphisms so long as the stated exact block or field-action data are recomputed at the actual period.

Costs and boundaries. Integer-basis completeness and integral witnesses are essential; rational saturation can introduce unsupplied obligations. Basis entries, interpolation denominators, and projector norms may be large. No uniform numerical conditioning in N, d or cycle heights is proved. The full-period seed, semiregularity, spin character, Hilbert spread, and isogeny descent remain geometric inputs. The finite library does not prove the Hodge conjecture on its own powers. A positive numerical minor cannot create that input. We have not independently audited the whole public manuscript or the private corpus.

Feasible extensions. Optimize an integral basis using the weighted slot budget, retaining exact unimodular certificates; compare it with Hilbert atoms when a specific supply construction is cheaper than cancellation. Record the actual CM witness library in applications, then use any independently established all-powers theorem available for that particular library. For higher CM fields, replace the two-endpoint calculation by a complete Galois-orbit spectral projector, with explicit collision analysis. For numerical work, evaluate the polynomial correspondence through certified factorizations and physical mass matrices, retaining every denominator and operator error. None of these proposals is presented as a completed new geometric Hodge theorem.

9 Contribution Assessment

C3 is warranted by the proved source-specific group reduction of supply obligations, the finite-library localization, and the constructive full-Weil-plane projector and correlated error certificate. The Hilbert-basis lemma, integer normal forms, invariant theory, and the qualitative one-class-to-plane implication are classical or already in the source, and receive no independent novelty credit. No C4/C5 claim is made: we produce no previously unknown cycle, remove no unproved geometric seed, and establish no new unconditional global Hodge case. The source applicability remains L3.

References

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